Industrial Telemetry Downtime Analysis using Tableau

Interactive Dashboard Project

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Executive Summary

This report documents the design and implementation of an interactive Tableau dashboard created to analyze industrial machine downtime across four Daikibo manufacturing facilities. Using telemetry data collected during May 2021 at 10-minute intervals, the project answers two business questions: which factory experienced the highest downtime, and which machine categories contributed the highest downtime within that factory.

The telemetry data was imported into Tableau, and a custom calculated field named ’Unhealthy’ was defined to represent the 10-minute intervals of downtime. Using this calculated field, two visualizations were created: a factory-level downtime bar chart and a machine-type downtime bar chart. The visualizations were combined into an interactive dashboard, which is configured with filter actions. Selecting a factory dynamically updates the machine category visualization.

The deliverable is a functional, interactive dashboard that allows immediate operational insight. This report details the tools, methodology, design, and observations, serving as an industry-standard technical showcase for data analytics portfolios.

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1 Introduction

Modern manufacturing operations rely heavily on telemetry monitoring to maintain equipment health and prevent unexpected production stops. This report documents the analysis of telemetry data collected from Daikibo's international manufacturing network. By visualizing machine state intervals, this project provides a clear operational view of equipment performance across multiple global facilities.

2 Business Problem

The client requested an analysis of industrial telemetry data to answer two specific operational questions:

1. Which factory experienced the highest downtime?
2. Which machine categories contributed the highest downtime within that factory?

The scope of work is strictly limited to answering these questions using the provided telemetry dataset.

3 Dataset Overview

The analysis is based on a JSON-formatted telemetry dataset covering the period of May 2021. The dataset contains telemetry messages collected from four manufacturing facilities:

- Daikibo Factory Meiyo (Tokyo, Japan)
- Daikibo Factory Seiko (Osaka, Japan)
- Daikibo Berlin (Berlin, Germany)
- Daikibo Shenzhen (Shenzhen, China)

Each location contains nine machine types. Telemetry messages are generated every 10 minutes.

4 Tools Used

The analysis and dashboard construction were performed using the following resources:

- Tableau: Used for importing data, defining metrics, creating visualizations, and designing the interactive dashboard layout.
- JSON Telemetry Dataset: The raw data source containing industrial logs from May 2021.

5 Methodology

The development workflow was executed in a structured path:

1. JSON Telemetry Import: Imported the raw JSON telemetry dataset into Tableau, establishing the structured fields for analysis.
2. Calculated Field Creation: Created a single calculated measure field named 'Unhealthy'. This field represents 10 minutes of downtime, which corresponds to the telemetry interval.
3. Factory Downtime Sheet (Sheet One): Developed a bar chart visualization displaying total downtime per factory to identify the location with the highest operational impact.
4. Machine Downtime Sheet (Sheet Two): Developed a bar chart visualization displaying downtime per machine type to isolate specific category contributions.
5. Interactive Dashboard Assembly: Combined Sheet One and Sheet Two into a single cohesive dashboard layout.
6. Filter Action Configuration: Configured a dashboard filter action where selecting a factory in Sheet One automatically filters the machine-type visualization in Sheet Two.

6 Dashboard Design

The dashboard design consists of two main sheets organized to show high-level factory metrics and low-level machine details. The first sheet, ’Downtime Per Factory’, displays a bar chart of total downtime for the four factories. This sheet enables immediate visual comparison to identify the factory with the highest total downtime. The second sheet, ’Downtime Per Machine Type’, is a bar chart representing downtime for each of the nine machine types.

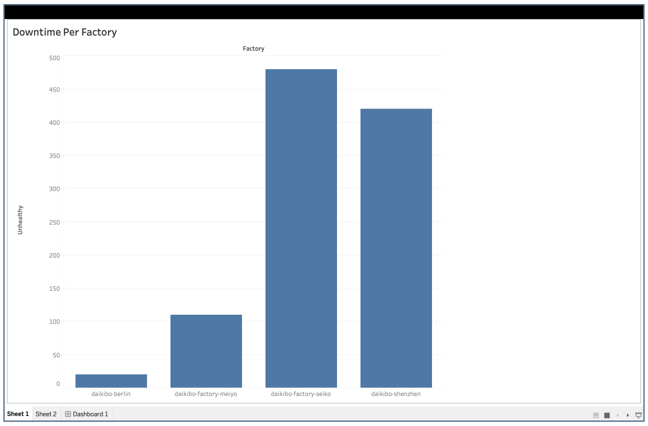

Figure 1 - Factory Downtime Visualization

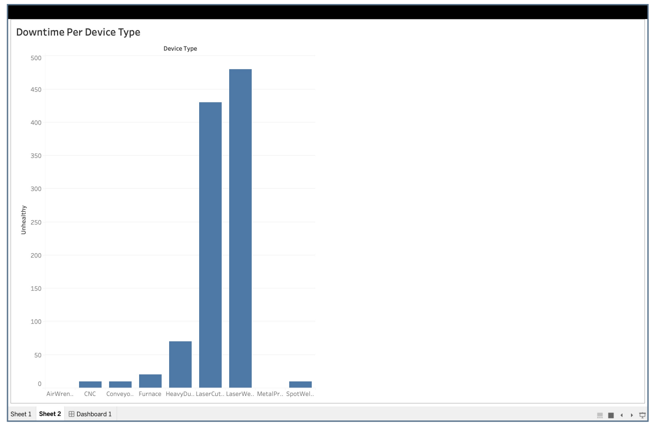

Figure 2 - Machine Downtime Visualization

7 Dashboard Walkthrough

The dashboard is designed for interactive exploration. When a user clicks on a factory bar within Sheet One (e.g., Daikibo Factory Meiyo, Daikibo Factory Seiko, Daikibo Berlin, or Daikibo Shenzhen), Tableau executes the filter action. This interaction dynamically updates Sheet Two to display only the machine types in that selected factory. The user can then inspect the filtered bar chart in Sheet Two to identify the machine categories responsible for the downtime at that specific factory.

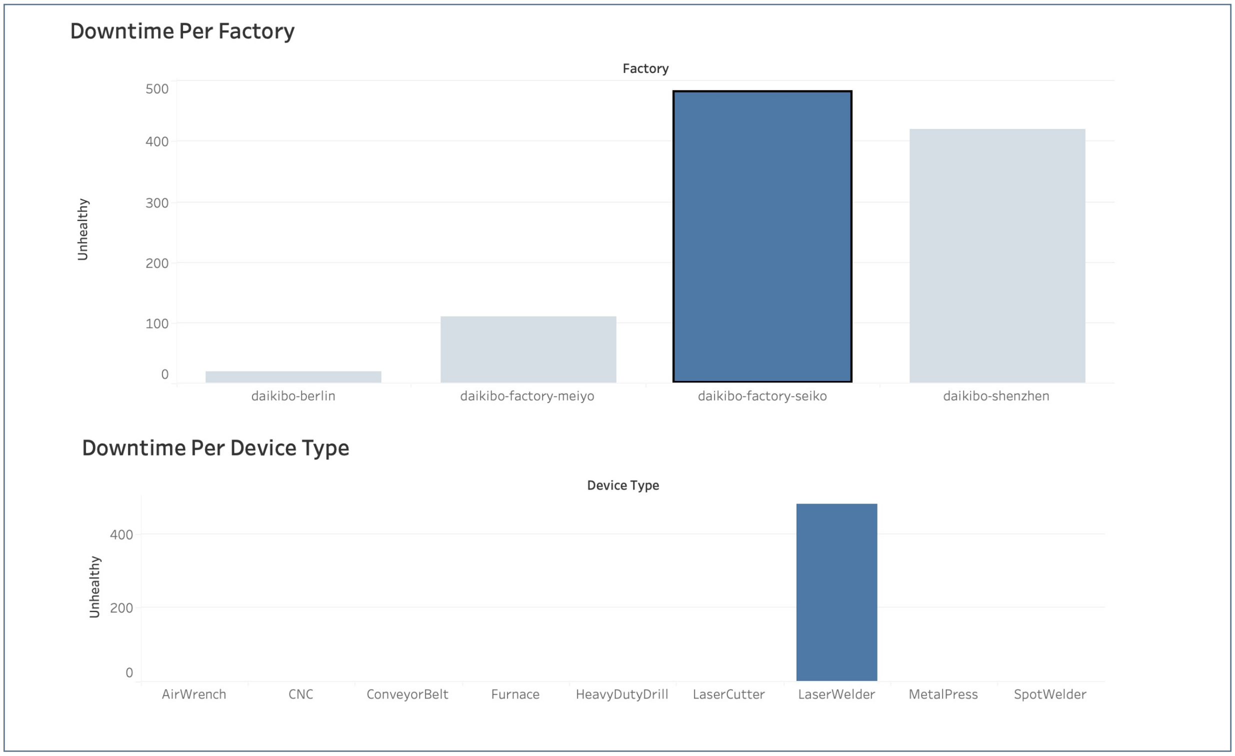

Figure 3 - Interactive Dashboard

8 Observations

The Tableau dashboard enables users to perform the following analysis:

- Compare overall downtime across all four Daikibo manufacturing locations (Tokyo, Osaka, Berlin, and Shenzhen).
- Identify which specific factory logged the highest cumulative downtime during May 2021.
- Drill down into any selected factory to examine the downtime contributions of its nine machine types.
- Isolate the primary machine categories responsible for the highest downtime within the highest-downtime factory.

9 Skills Demonstrated

- Tableau Desktop: Data visualization and dashboard construction.
- Calculated Fields: Creating operational measures (e.g., ’Unhealthy’ downtime intervals).
- Data Import: Importing JSON telemetry logs.
- User Interaction Design: Configuring dashboard actions and interactive filtering.
- Portfolio Project Presentation: Structuring and documenting business-focused technical reports.

10 Conclusion

The Tableau dashboard successfully addresses the client’s business objectives. By importing the JSON telemetry dataset and implementing the ’Unhealthy’ calculated measure, the dashboard provides clear answers to the factory and machine downtime questions. The interactive filter action enables immediate operational analysis, making this project a valuable addition to a professional data analytics portfolio.